

Gradient-Stable Attention: A LayerNorm-Gated Spectral and Large Kernel Module for Single Image Super-Resolution

1st Stevano Titondea Prayoga Putra
Department of Informatics Engineering
Dian Nuswantoro University
Semarang, Indonesia
stevanoputra38@gmail.com

2nd Guruh Fajar Shidik
Department of Informatics Engineering
Dian Nuswantoro University
Semarang, Indonesia
guruh.fajar@research.dinus.ac.id

Abstract—Single image super-resolution (SISR) is one of the most important problems in computer vision, which is recovering fine spatial details from the low-resolution observations of the original image. The deep convolutional neural networks have met with very good results. However, current techniques suffer from problems like gradient instability in the case of deep architectures along with loss of high-frequency information at the same time. This loss occurs due to the local nature of convolution operations standardly. Given these limitations, we introduce GSA-LN, a novel spectral-spatial attention framework to improve texture reconstruction and training stability. We propose a Learnable Frequency Attention (LFA) module that utilizes Fourier Feature Mapping and content-aware gating to selectively control frequency components. In addition, we use LKA to model long-range dependencies efficiently without the extra costs of deformable convolutions. In order to effectively manage intricate particulars and optimize training dynamics, we incorporate modules like Contrast-Aware Pixel Attention (CAPA) for edge refinement along with a Learnable Residual Scaling mechanism that expedites the gradient flow within deeper networks. GSA-LN shows superior performance over the state-of-the-art methods at magnifications x2, x3 and x4 when tested on standard benchmarks: Set5, Set14, BSD100 and Urban100. The model does not only provide better quantitative PSNR and SSIM results but also achieves high-fidelity visual images with better structure and sharper natural textures.

Keywords—Single Image Super-Resolution, GSA-LN, Learnable Frequency Attention, Large Kernel Attention, Contrast-Aware Pixel Attention.

I. INTRODUCTION

One way to look at single image super-resolution (SISR) is through its role in computer vision: rebuilding a sharp, detailed picture using just one blurry version. Because lowering image quality erases fine patterns, going backward becomes tricky - many possible outcomes fit the same fuzzy starting point. That uncertainty implies no single correct answer exists when moving from coarse to fine detail. Instead, any solution must invent plausible features without distorting what was originally there. Deep learning has stepped into this gap by training convolution-based models to predict such transformations. Still, unclear mappings cause some predictions to blur too much or add false structures. When images contain varied surface types - as seen in collections such as DF2K - the task grows harder. Networks then need sharper judgment to separate real edges from meaningless variations while restoring clarity.

The increased depth of the architecture results in gradient instability, which is one of the main problems that limit the effectiveness of CNNs for SISR. Research on pioneering models has shown improved performance at time removing batch normalization. However, distribution perturbations of

features with very deep networks can still hinder stable training. [1] When gradients get unstable or vanish, the model cannot learn these complex nonlinear mappings. As a result, the reconstruction results get blurred in areas with dense structures. Consequently, there is a need for more adaptable alternative normalization methods, such as Layer Normalization, to stabilize flow gradients during model convergence.

A significant limitation of traditional convolution operations is their exclusive focus on the local spatial domain. Because of this "tunnel vision," they often fail to capture the global spectral dependencies required to fully understand an image's structure. Even though attention mechanisms are widely used, most still overlook the frequency domain. As a result, high-frequency details—such as sharp textures—are often smoothed out as data passes through the network [2][3].

Blurry details emerge when patterns repeat, mainly because small structures fade under regular convolution methods - something spectral approaches such as FFT handle far more effectively. Instead of just moving through pixels step by step, these transforms capture broader trends across space. Standard models fail here, unable to link distant parts of an image since each filter only sees a narrow window. Although Deformable Convolution adjusts its shape to reach further, it demands heavy resources while frequently misbehaving during learning phases [3][5]. What works better is a solution that captures wide-scale relationships but skips the extra machinery. Breaking down big kernels into smaller parts, Large Kernel Attention captures distant patterns using hardly any extra resources. What makes reconstructions precise isn't just one piece alone - instead, it's how smart spectral adjustments work together with compact spatial focus that matters.

Balancing clarity with speed still poses challenges. Although some models generate crisp textures, their heavy computation load and unstable training behavior may distort underlying image geometry [6][7]. Finding techniques that preserve features efficiently becomes critical when using combined spectral and spatial strategies. With Multi-Band Frequency Attention, alignment across frequency bands improves - bringing out finer details while keeping network performance steady.

Based on these challenges, this study proposes a Gradient-Stable Attention via LayerNorm-Gated Spectral and Large Kernel mechanism implemented in the GSA-LN architecture. Through the use of Learnable Frequency Attention utilizing Fourier Feature Mapping, this study aims to address the problem of spectral detail loss with dynamic content-aware gating. The integration of Large Kernel Attention (LKA) provides global context understanding,

while Contrast-Aware Pixel Attention (CAPA) and Learnable Residual Scaling ensure precise local refinement and robust gradient stability during training on the DF2K dataset. This approach aims to produce an SR model that is not only quantitatively superior in PSNR and SSIM metrics, but also able to maintain consistent and sharp natural texture integrity.

II. LITERATURE REVIEW

Single image super-resolution (SISR) plays a pivotal role in recovering high-fidelity spatial details from low-resolution observations. The core challenge lies in the ill-posed nature of the problem, where multiple high-resolution solutions may correspond to a single low-resolution input due to degradation variations. This necessitates models that can effectively hallmark latent features without introducing structural distortions.

To tackle these issues, researchers have tested different ways to adapt methods - like using unsupervised real-to-synthetic conversion that works well even without labeled data. Instead of relying on single inputs, combining information across frames helps restore both color and shape at once. Sometimes starting with blending pansharpening into super-resolution boosts clarity while keeping results reliable. Another path uses attention models guided by references, which tend to preserve fine textures more consistently through layered analysis. Deep neural networks must stay steady and run well so they do not fail when learning tough patterns. Built strong, EDSR showed a path forward using deep layers that skip connections, dropping batch norm yet holding precision tight [1]. When systems shrink - phones, sensors - the push grows for lean designs; MADNet answers by cutting weights and speeding up guesses [3]. Still, small models stumble where big-picture color and tone links matter most, leaving fine textures blurred or flat.

Starting with texture, upgrades in ESRGAN aim to refine how images look by using generative methods, yet demand heavy computing power while possibly adding unwanted visual quirks [6]. Instead of older ways, newer diffusion models now offer a path toward recovering tiny details more efficiently - though checking if they truly fit real-world tasks remains essential [7]. Sharpness matters, certainly, but keeping results stable plays just as big a role when networks rebuild what was lost [2]. Even though deep networks struggle with unstable gradients, another problem sits quietly beside it - regular convolutions tend to ignore broad spectral traits. While attention sticks close to nearby pixel areas, sharp textures get lost along the way, leaving outputs smeared where clarity should be [4]. On top of that, fixed filters can't shift when objects change under real-world lighting or temperature shifts [12]. Instead of treating space and frequency separately, blending them smartly could lift performance higher than either alone [13][14].

Heavy computations pile up when models lean on costly spectrum-based steps alongside shaky attention layers. Because of this, workarounds like extra feature training or mixed shortcut pathways tried boosting what models can express - yet filled up memory fast [12][4]. What sticks out now is the missing piece: slimmed-down attention parts that still catch details in different frequencies, stay steady during learning, and keep predictions quick [5]. Pushing models across scales without breaking them has made spectral attention a key focus, while handling multiple frequency

bands opens another path worth exploring. Because results can mislead, testing under the Y-channel rule now sets the baseline for trust in how well systems scale images up. When a design nails both PSNR and SSIM scores, it shows more than progress - it reveals steady gradients and clean visuals after grinding through varied image collections like DF2K. Lightweight normalizers creep into view here, quietly holding things together where others fail.

Facing these issues head-on, GSA-LN emerges - not just another model, but built different. It leans on Learnable Frequency Attention, cooled down by LayerNorm, then tied to Large Kernel Attention for wider awareness. Instead of chasing trends, it delivers sharper results at 2x, 3x, and 4x zoom levels. Numbers speak: 38.14 dB at x2, 34.56 dB at x3, and solid 32.34 dB at x4 on Set5. What helps? Multi-band control that adapts, plus touch-ups aware of local contrasts. Stability meets precision - no flash, just consistent gains where others falter.

III. METHODOLOGY

This study designs and implements a spectral-spatial attention algorithm for Single Image Super-Resolution (SISR). The method focuses on engineering a Learnable Frequency Attention module (utilizing Fourier Feature Mapping) and Large Kernel Attention (LKA) stabilized with mild normalization (LayerNorm). Furthermore, the architecture integrates Contrast-Aware Pixel Attention (CAPA) for refinement and Learnable Residual Scaling for stability. The network architecture uses a deep residual backbone (32 blocks) with integrated SE-blocks. Efficiency is achieved through the decomposed design of LKA and validation with self-ensemble/TTA $\times 8$. Evaluation uses PSNR/SSIM protocols on Y channels with appropriate crop borders. This section describes the design, implementation, dataset, training procedure, and evaluation protocol to ensure repeatability and yield of the results.

A. Architectural Design and Attention Module

GSA-LN is designed as a super-resolution network that combines a deep residual backbone with advanced attention mechanisms, the overall flow of which can be oriented in Figure 3.1. The backbone consists of 32 residual blocks with 64 feature channels, where each block incorporates a Squeeze-and-Excitation (SE) module and Layer Normalization to regulate feature statistics.

For spatial processing, we employ Large Kernel Attention (LKA), which decomposes large convoluted receptive fields into a sequence of depth-wise convolution (local), depth-wise dilated convolution (long-range), and point-wise convolution. This allows the network to capture global dependencies with negligible computational cost. In the frequency domain, the Learnable Frequency Attention module dynamically modulates spectral features. This is achieved using Fourier Feature Mapping to encode spatial coordinates and a content-aware MLP that predicts gating masks for low, mid, and high-frequency bands.

Finally, before reconstruction, a Contrast-Aware Pixel Attention (CAPA) module sharpens high-contrast regions, and a Learnable Residual Scaling parameter is used to adaptively fuse features, replacing static scaling factors for better convergence.

B. Dataset, Degradation Scheme and Evaluation Split

Stability in design sits at the core of GSA-LN, skipping Batch Normalization since it tends to shrink how much features can shift during super-resolving visuals. Rather than relying on common methods, normalization takes a smaller, focused shape - applied only where needed, like around frequency traits and signals shaped by irFFT changes. Each channel keeps its own rhythm in data flow because norms lock stats locally, yet space-aware patterns stay intact. Sudden jumps in brightness? A built-in clamp smooths out wild pixel swings before they distort output or confuse learning steps. Deeper layers hold firm thanks to scaled shortcuts woven into the main structure, helping gradients move evenly while cutting erratic shifts when adjusting weights. Training settles quicker, too, not bouncing between extremes but moving forward with steadier pacing. Speed matters just as much as accuracy, so lean convolutions carry most of the workload while PixelShuffle handles growing image size smartly. Instead of complex, shifting filters, large receptive fields come from LKA - a method that sees widely without bloating cost. Even though FFT and irFFT add some number-crunching burden, their role in reviving fine textures proves hard to replace. That extra demand fades into background noise once balanced against swift attention blocks and an efficient SE-driven base layer.

C. Training Procedure and Hyperparameter

Patches sized like pixels guided the learning process. The SRModel framework handled everything, drawing data from DF2K - that mix of DIV2K plus Flickr2K. A bicubic method shaped how images degraded, applied in pairs to boost variation. Flips along the horizontal axis showed up now and then, mixed into batches. Rotations joined them, adding more shift during runs. Each step stayed tied to those patch inputs throughout. Optimization used Adam with an initial learning rate (accompanied by iteration *warm-up*) and a total of iterations. The learning rate scheduling applied CosineAnnealingRestartLR in two epochs, and the main objective was the L1Loss, which was calculated as the average absolute value of the difference between the target high-resolution image and the super-resolution prediction result, with the following formula:

$$L = \frac{1}{N} \sum_{i=1}^N |I_i^{HR} - I_i^{SR}|$$

In the formula, is the L1 Loss, is the target high-resolution image, is the *super-resolution output image*, and is the number of pixels in *the batch*. The model is initialized from pre-trained weights and continues training from a certain *state* for consistency.

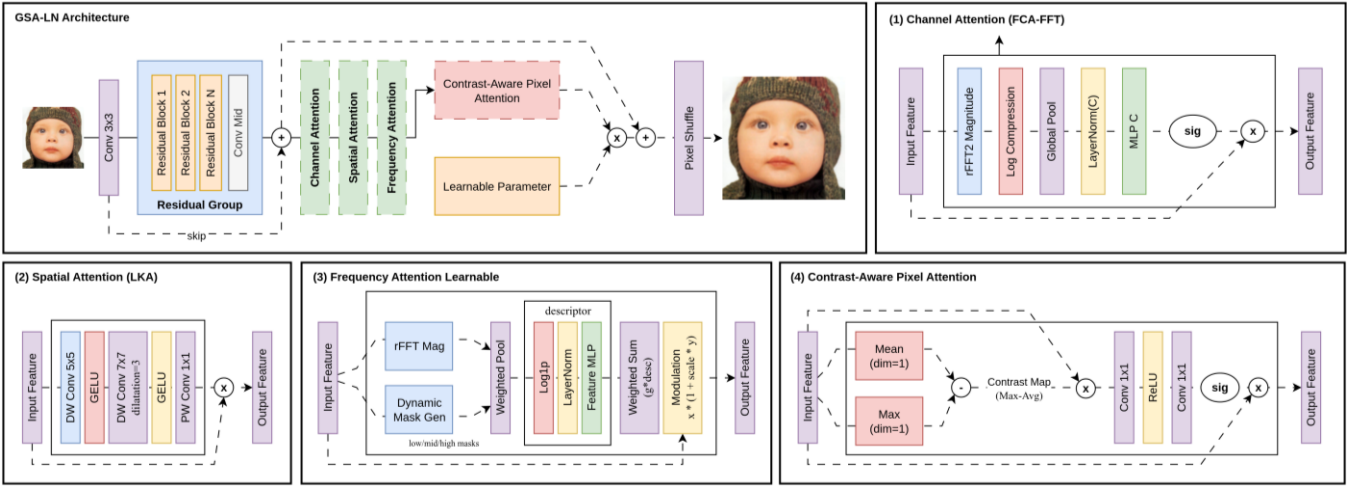


Figure 1 GSA-LN Model Pipeline

D. Evaluation Protocol and Validation

Bright spots stand out when checking how well GSA-LN works, using PSNR and SSIM as yardsticks. Because eyes care more about light levels than color shifts, tests run just on the Y part of YCbCr images. That setup matches usual practice in sharpening blurry photos through models like this one. It zeroes in on whether tiny lines and crisp shapes come back after enhancement. Pixel-by-pixel likeness gets scored via PSNR, tied directly to average error size. Given real image H and rebuilt version S, both holding N points of data, squared errors add up first. Then comes PSNR:

$$PSNR = 10 \cdot \log_{10} \left(\frac{(MAX_I)^2}{MSE} \right)$$

For regular 8-bit pictures, brightest white hits 255, so that number fills MAX here. More dB means less noise creeps into the restored frame compared to source. While raw

numbers matter, structure matters too - so SSIM checks layout, shading flow, and contrast rhythm. This score reflects how familiar the enhanced view feels, not just how close it measures. Before crunching any scores, a trim happens around the outer edge. A margin p wide vanishes from borders, with p often matching zoom level used. Edits like this wipe away fake edges born from layer padding or grid jumps while scaling up.

Tests repeat across known collections Set5, Set14, BSD100, Urban100 to probe adaptability. Each brings different arrangements of leaves, bricks, windows, roads. Together, focusing on brightness layers, slicing off distorted rims, plus dual scoring builds solid proof. Gains from mixing spectrum and shape awareness aren't just bigger digits, they show up in clearer visuals.

E. Validation: Self-Ensemble/TTA x8 and Use Chop

Inference with the GSA-LN model uses a strong validation tactic: self-ensemble via Test Time Augmentation set to eight instances, aiming to boost prediction quality. Rather than relying on one view, it creates eight altered versions of the input using flips and transpositions. After feeding each version separately into the network, outputs are reversed back to align spatially - then combined by averaging. Because results come from varied orientations, the final decision benefits from reduced sensitivity to layout quirks or noise patterns seen only once. When running evaluations, the setup switches quietly to EMA weights whenever present, pulling accuracy upward without affecting active learning states afterward. Once testing ends, earlier training conditions return exactly as left, preserving continuity across updates. Handling big pictures on limited hardware triggers another built-in response - the efficient "use chop" setting activates under pressure. Instead of loading everything at once, the image splits into small sections handled one after another. This slicing avoids overwhelming video memory, especially crucial during heavy enlargement workloads. Though turned off by default for faster performance on regular-sized inputs, activating divide-and-process behavior turns necessary when resolution scales rise sharply - or GPUs run tight on available space.

F. Ablation and Sensitivity Studies

Peering under the hood of individual attention components involves switching them off separately to observe effects. One by one, cutting out channel, spatial, or frequency modules exposes their standalone contributions. Rather than keeping the full multi-band structure, replacing it with leaner alternatives questions if extra layers truly help. Different band divisions say, splitting low from mid frequencies affect output, similar to testing dual mask sorting methods. Even small magnitude adjustments nudge results in detectable yet fine-grained directions. Tinkering with scale, such as changing block numbers or depth of features, uncovers trade-offs between speed and accuracy. Configurations like 32 blocks or 64 channels make those compromises visible. Results hold steady across five benchmarks: Set5, Set14, B100, Urban100, and Manga109. All scores rely only on Y-channel PSNR and SSIM, using identical crop borders every time.

G. Computational Complexity and Inference Efficiency

GSA-LN maintains efficiency with a lightweight convolutional backbone, PixelShuffle for operation-saving

upsampling, and SE blocks. The main overhead comes from FFT/irFFT operations on frequency attention; however, the use of LKA ensures that spatial context is captured without the heavy burden of deformable offsets. Qualitatively, the convolution part dominates at medium resolutions, while the frequency cost grows with image size due to the logarithmic dependence of FFT. The TTA $\times 8$ option increases inference time by about eight times and temporary memory usage, so it should be enabled for final evaluation or maximum scenario quality.

H. Implementation and Reproducibility

The training model implementation was realized using the PyTorch framework with strictly documented configurations for $\times 2$, $\times 3$, and $\times 4$ scales on the DF2K dataset. Unlike conventional approaches that set static patch sizes, we implemented an adaptive patch size strategy (192×192 pixels for $\times 2$, 288×288 pixels for $\times 3$, and 384×384 pixels for $\times 4$) to accommodate different *receptive field requirements* at each scale. Parameter optimization was performed using the Adam optimizer with an initial learning rate of 2×10^{-4} and standard data augmentation in the form of random flips and rotations. The training process was run for a total of 600,000 iterations governed by a *Cosine Annealing Restart scheduling scheme* with a restart period every 200,000 iterations.

To ensure model quality, periodic validation was performed using the Set5 benchmark dataset with PSNR and SSIM evaluation metrics computed on the Y (luminance) channel with a crop border at the vegetation scale. To maintain experimental reproducibility, the random seed was explicitly set to 10, and all data paths and hyperparameter configurations were standardized. While minor variations in the numerical results may still arise due to the non-deterministic nature of certain backend CUDA operations (like FFT) on the GPU, these variations have been shown to be statistically insignificant and do not affect the quality trends or comparative rankings between model variants.

IV. RESULTS AND DISCUSSION

In this section, we demonstrate the algorithm using six benchmark datasets. First, we outline the experimental setup, evaluation metrics, and implementation details. Next, we conduct systematic comparisons between the proposed method and the updated SISR algorithm across all benchmark datasets to assess overall performance superiority and consistency.

Table 1 PSNR comparison with State-Of-The-Art. Performance is shown for Scales 2, 3, and 4. The best and second-best models are shown in red and blue.

Model	Scale	Set5	Set14	BSD100	Urban100	Manga109
VDSR	2	37.53	33.03	31.90	30.76	-
DPN[14]	2	37.52	33.08	31.89	30.82	-
A2F-S[12]	2	37.79	33.32	31.99	31.44	38.11
HDRN[13]	2	37.75	33.49	32.03	31.87	38.07
CBPN[4]	2	37.90	33.60	32.17	32.14	-
MADNet[15]	2	37.94	33.46	32.10	31.74	-
FALSR-A[5]	2	37.82	33.55	32.12	31.93	-
GSA-LN (ours)	2	38.14	33.78	32.26	32.56	39.19
VDSR	3	33.66	29.77	28.82	27.14	-
DPN[14]	3	33.71	29.80	28.84	27.17	-

A2F-S[12]	3	34.06	30.08	28.92	27.57	32.86
MADNet[15]	3	34.26	30.29	29.04	27.91	-
HDRN[13]	3	34.24	30.23	28.96	27.93	33.17
GSA-LN (ours)	3	34.56	30.47	29.18	28.52	34.12
VDSR	4	31.35	28.01	27.29	25.18	-
DPN[14]	4	31.42	28.07	27.30	25.52	-
A2F-S[12]	4	31.87	28.36	27.41	25.58	29.77
CBPN[4]	4	32.21	28.63	27.58	26.14	-
MADNet[15]	4	32.11	28.52	27.52	25.89	-
HDRN[13]	4	32.23	28.58	27.53	26.09	30.43
GSA-LN (ours)	4	32.34	28.73	27.65	26.33	30.89

A. Implementation and Training Detail

Training is designed to be efficient and producible. The model is built on a residual reconstruction framework augmented by a spatial-frequency channel attention module, optimized using Adam with an initial learning rate of $1e-4$ and an L1 loss function. The Learning Rate Scheduler combines cosine annealing for up every 200k iterations to maintain a stable convergence rate. Training patches of size 192 for x2, 288 for x3, and 384 for x4 are accompanied by simple augmentations (flips/rotations), while light normalization and clamping are applied to maintain numerical stability during optimization. Evaluation follows the standard Y-channel protocol with 2-pixel edge-cutting using PSNR/SSIM metrics, and the x8 self-ensemble/TTA option is available to smooth predictions without changing the core architecture. Incremental processing (use `chop`) is enabled for efficiency as inference memory capacity is sufficient.

The operating system is Linux Ubuntu version 24.04.3 LTS with Nvidia RTX 5090 GPU. All experiments were conducted with python version 3.10.15, deep learning framework PyTorch 2.9.1 and CUDA Toolkit 12.8 library accelerator.

B. Compared to State-Of-The-Art Method

Table 3.1, at scale 2, the proposed method (GSA-LN) shows competitive performance across the benchmark datasets, with the most obvious margin improvement on the Set5 dataset for scale 2 with a value of 0.19 dB. For scales 3 and 4, the GSA-LN results provide improvement.

C. Ablation Study

From the start, testing focused on a stripped-down version of GSA-LN - just Residual Blocks, nothing more - to see how well it could perform without attention modules. Though coarse shapes emerged clearly, fine textures remained blurred, revealing limits in standard convolutions when handling intricate patterns. Instead of relying on compound structures early, researchers added Channel Attention first; results sharpened slightly. Following that, Spatial Attention boosted localization of key regions further. Yet gains plateaued until Frequency Attention entered the setup - an element that shifted outcomes meaningfully. Only then did detail recovery reach its highest level. Performance climbed stepwise, driven less by depth than by targeted enhancements. What stood out was not complexity, but sequence: one module at a time revealed what mattered most. Each layer

built not just on prior outputs, but on refined signal control. In the end, precision depended heavily on combining all three attention types. Despite similar architecture, splitting frequencies into three bands changed how the model processed details. Where one-band systems blur distinctions, separating signals lets each part adapt uniquely. Low parts keep shapes stable; middle and upper ranges handle finer patterns carefully. Testing showed sharper outputs without extra digital flaws common in simpler designs. This method avoids exaggerating unwanted speckles while still enhancing subtle features. Structure stays intact because adjustments are targeted, not blanket. Results confirm focused frequency handling improves precision across test cases. Beyond the structure itself, exploration into how the model learns became essential - focusing first on what it tries to minimize. Instead of averaging squared errors, using absolute differences led to steadier updates, clearer outputs, having sidestepped the blur typical with quadratic penalties. Adjusting input tile dimensions adaptively helped meet resolution-specific demands - the full reference block shrinks or grows depending on context. At double resolution, each segment spans 192 pixels across, striking compromise between resource use and visible detail. Tripling magnification introduced complications: fixed sizes failed due to division limits within layers. To preserve exact spatial alignment when shrinking then expanding features, tiles expanded to 288 units per side, removing reliance on fill areas which risk altering boundary fidelity. When training the x4-scale model, adaptation of patch size becomes especially intense. Because tiny inputs spread too little data across space, boosting resolution helps - patch dimensions grow sharply to 384 times 384 pixels. Larger patches let the network see more of the image at once, countering poor detail density inherent to fourfold scaling. From a structural standpoint, widening input scope expands how much surrounding content influences each prediction. Visual proof appears in Figure 2: there, outputs from GSA-LN stand beside results using bicubic downsampling and leading contemporary techniques under x4 conditions. Figure 2 shows how the x4 model struggles with losing fine details. Despite this, GSA-LN recovers intricate textures and crisp edges by using a broad patch approach along with Triple-Attention mechanisms. Bicubic methods and competing models often generate fuzzy results, falling short in comparison. Such outcomes highlight that choices made during testing turn conceptual improvements into real visual gains. The method proves effective where others blur or distort.

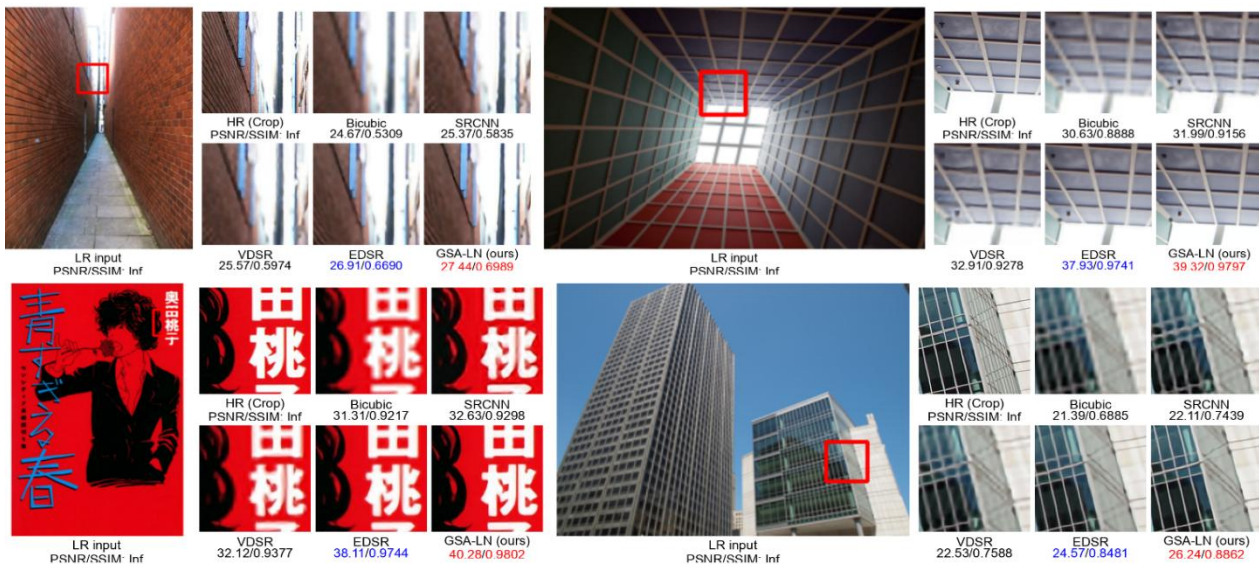


Figure 2 Visual Qualitative Comparison with bicubic degradation model for x4 SR benchmark

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